A TENSOR-PRODUCT POSITIVE SUBCONE IN THE $L^p(L^q)$ DILATION PROBLEM

AGENT LANE 04

Status

This packet gives a positive partial result for Open Problem 7.6 in Fackler–Gluck, *A Toolkit for Constructing Dilations on Banach Spaces* [1]. The open problem asks for the weak operator closed convex hull of the positive invertible isometries on $L^p([0,1]; L^q([0,1]))$. We prove that this hull contains all spatial tensor products of scalar positive contractions, and hence the weak operator closed convex hull generated by finite convex combinations of such tensor products.

Source open problem

The source asks:

> Let $p, q \in (1, \infty)$ and let $\mathcal{T}$ denote the weak operator closure of the convex hull of all positive invertible isometries on $L^p([0,1]; L^q([0,1]))$. Does $\mathcal{T}$ coincide with the set of all positive contractions? If not, can a good characterization of the elements of $\mathcal{T}$ be given?

22 STEPHAN FACKLER AND JOCHEN GLÜCK

Open Problem 7.6. Let $p, q \in (1, \infty)$ and let $\mathcal{T}$ denote the weak operator closure of the convex hull of all positive invertible isometries on $L^p([0,1]; L^q([0,1]))$. Does $\mathcal{T}$ coincide with the set of all positive contractions? If not, can a good characterization of the elements of $\mathcal{T}$ be given?

Dilations on Hilbert spaces. In the previous subsection we considered a dilation

Idea of the proof

The scalar $L^r[0,1]$ case is already available in the source: positive contractions are weak operator limits of convex combinations of positive invertible isometries. If U is a positive invertible isometry on the outer L^p space and V is one on the inner L^q space, then the spatial operator $U \otimes V$ is a positive invertible isometry on the mixed-norm space. Taking convex combinations and then weak operator limits on algebraic tensor test functions transfers the scalar approximation into the mixed space.

Partial result

Let G_r denote the set of positive invertible isometries of $L^r[0,1]$. Let

$$X_{p,q} = L^p([0,1]; L^q([0,1])).$$

Let $\mathcal{T}_{p,q}$ be the weak operator closure in $\mathcal{L}(X_{p,q})$ of the convex hull of the positive invertible isometries of $X_{p,q}$.

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Theorem 1. *Let $1 < p, q < \infty$. If A is a positive contraction on $L^p[0,1]$ and B is a positive contraction on $L^q[0,1]$, then the spatial tensor product $A \otimes B$ belongs to $\mathcal{T}_{p,q}$.*

Consequently, $\mathcal{T}_{p,q}$ contains

$$\overline{\mathrm{conv}}^{\mathrm{WOT}}\{A \otimes B : A \geq 0,\ \|A\|_{L^p \to L^p} \leq 1,\ B \geq 0,\ \|B\|_{L^q \to L^q} \leq 1\}.$$

Proof. The source proves, using Greim's theorem, that every positive contraction on $L^r[0,1]$ lies in the weak operator closure of $\mathrm{conv}(G_r)$ for $1 < r < \infty$; see the proof of Lemma 7.2 in [1]. Choose nets

$$A_i \in \mathrm{conv}(G_p), \qquad B_j \in \mathrm{conv}(G_q)$$

with $A_i \to A$ weakly on $L^p[0,1]$ and $B_j \to B$ weakly on $L^q[0,1]$.

First suppose $U \in G_p$ and $V \in G_q$. The operator $U \otimes V$, initially defined on algebraic tensor functions by

$$(U \otimes V)(a \otimes b) = Ua \otimes Vb,$$

extends to a positive invertible isometry of $X_{p,q}$. Indeed, positive surjective isometries of scalar L^r spaces are lattice isomorphisms; the inner isometry $I \otimes V$ preserves the L^q norm in each fibre, and the outer isometry $U \otimes I$ preserves the outer L^p norm. Thus their composition $U \otimes V$ is a positive invertible isometry on $X_{p,q}$.

It follows that, for each pair (i, j),

$$A_i \otimes B_j \in \mathrm{conv}\{U \otimes V : U \in G_p,\ V \in G_q\} \subseteq \mathcal{T}_{p,q}.$$

All these operators are contractions on $X_{p,q}$.

We now check the weak operator limit. For algebraic tensors $a \otimes b \in L^p \otimes L^q$ and $a' \otimes b' \in L^{p'} \otimes L^{q'}$, where p' and q' are conjugate exponents,

$$\begin{aligned} \langle (A_i \otimes B_j)(a \otimes b), a' \otimes b' \rangle &= \langle A_i a, a' \rangle \langle B_j b, b' \rangle \\ &\longrightarrow \langle Aa, a' \rangle \langle Bb, b' \rangle. \end{aligned}$$

Finite sums give the same convergence for algebraic tensor test functions and algebraic tensor functionals. Since $1 < p, q < \infty$, the dual of $X_{p,q}$ is $L^{p'}([0,1]; L^{q'}([0,1]))$, and algebraic tensors are dense in both the space and its dual. Uniform boundedness of the contractions $A_i \otimes B_j$ therefore gives weak operator convergence to the bounded operator determined on algebraic tensors by

$$(A \otimes B)(a \otimes b) = Aa \otimes Bb.$$

Thus $A \otimes B \in \mathcal{T}_{p,q}$ because $\mathcal{T}_{p,q}$ is weak operator closed.

The final assertion follows because $\mathcal{T}_{p,q}$ is convex and weak operator closed. ☐

Limitations

This does not solve Open Problem 7.6. It only identifies a natural positive subcone contained in the desired hull. The result does not decide whether operators with genuinely non-product mixed kernels, or all positive contractions, lie in $\mathcal{T}_{p,q}$.

Novelty and verification notes

- Local run indexes were searched for `1709.08547`, the source title, `dilation`, `simultaneous dilation`, and `L^p(L^q)`. No exact prior packet for this arXiv id was found. Related dilation attempts for other papers were checked for overlap.
- Bounded web search on 2026-06-29 for the source title and the phrases "positive invertible isometries $L^p(L^q)$" and "weak operator closure positive contractions $L^p(L^q)$" found no later complete characterization.
- The proof uses only the scalar positive-contraction approximation already used in the source and standard density of algebraic tensors in $L^p(L^q)$ and its dual.

Human review recommendation

Review as a modest but likely valid partial result for Open Problem 7.6. The main point to check is the weak-operator tensor-limit passage; it is deliberately formulated on algebraic tensor tests to avoid relying on an unstated vector-valued extension theorem.

References

[1] S. Fackler and J. Gluck, *A Toolkit for Constructing Dilations on Banach Spaces*, arXiv:1709.08547, 2017. https://arxiv.org/abs/1709.08547